MG MUSALLAM GROUP · DELIVERY OPERATIONS

Firebase Account Transfer

Option A Runbook

How to move full ownership of the production Firebase / GCP project to a different Google account — IAM ownership, billing relink, credential rotation and rollback — with no new project, no data export and no application release.

Firebase / GCP | project musallam-delivery-prod | Zero downtime | Version 1.0 · 5 Nov 2026

Written from the project configuration, IAM and deployment scripts in this repository · English edition

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Moving full ownership of the Firebase / GCP project `musallam-delivery-prod` (project number `579224507592`) to a different Google account, **without creating a new project and without changing any application code**.

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1. What "transfer" actually means

Google Cloud has no project-transfer button. A project is a fixed object; only two things can change about who controls it:

1. **The IAM policy** — which Google accounts hold which roles. `roles/owner` is real ownership.
2. **The billing account link** — whose payment method is charged.

Everything else — project id, project number, Firestore data, Auth users, Cloud Functions URLs, API keys, FCM sender id — stays exactly as it is.

So "moving to another Google account" is: grant Owner to the new account, re-link billing, rotate credentials, then demote the old account.

Why Option A and not Option C

Option C (a brand-new Firebase project) is a different project with a different number. That forces a full data export/import, a staff password reset (Firebase-managed password hashes cannot be exported), a rebuilt Android app with a new `google-services.json`, a new Play Store release, and ~1,200 rider push tokens becoming unreachable until every phone upgrades. Option A avoids all of it.

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2. What does NOT change under Option A

Item	Why it matters
Project id and number (`579224507592`)	The FCM sender id is derived from the project number, so **no push token is invalidated**
Firestore data, indexes, TTL overrides	No export/import required
Firebase Auth users and custom claims	Staff session-cookie audience is unchanged — **no forced logout**
Cloud Functions URLs (`me-central1-…cloudfunctions.net`)	Worker, crons and the admin panel keep calling the same endpoints
Service account email and every API key	Nothing to re-issue except the ones you deliberately rotate
Vercel environment, Cloudflare Worker config, `.firebaserc` , `infra/`	**Zero code change in this repository**
Cloudflare R2 (`dpd-private-prod` , `dpd-gps-prod`)	Unrelated to Firebase; untouched
Play Store listing and rider APKs	No release needed

Option A is a permissions, billing and credentials operation. It is not a migration.

3. Prerequisites

Check	Consequence if missing
New Google account is ready (personal or Workspace)	—
You currently hold **Owner** on the project	Only an owner may grant Owner
You hold **Billing Account Administrator** on the target billing account	`billing.resourceAssociations.create` is denied
The target billing account already has a **payment method**	The relink fails without one
Workspace domain-restricted sharing permits the principal you are adding	Cross-domain `add-iam-policy-binding` is refused
The old account's 2FA recovery codes are saved	The original creator account cannot be recovered once locked out
The current billing account ID is recorded	Needed for rollback

> Do not delete the original creator account. Keep it as Viewer with an Essential Contact on it — Google still treats it specially for recovery.

4. Phase 1 — Snapshot for rollback

Run as the **current** owner.

```
gcloud auth login
gcloud config set project musallam-delivery-prod
gcloud projects get-iam-policy musallam-delivery-prod --format=json > iam-before.json
gcloud beta billing projects describe musallam-delivery-prod
```

Record `billingAccountName` and confirm `billingEnabled: true`. `iam-before.json` is the only exact rollback reference — keep it outside the repository.

Also capture the inventory while access is still known-good:

```
firebase functions:list      > functions-before.txt
firebase firestore:indexes   > indexes-before.txt
firebase auth:export staff-before.json
```

Pulumi state

The infrastructure stack in `infra/` reads its project id from `infra/Pulumi.production.yaml` (`dpd-infra:gcpProjectId`, default `musallam-delivery-prod`). Option A does not change that value. Confirm the stack secrets still resolve for the new account before any `pulumi up`:

```
cd infra
pulumi stack select production
pulumi config --show-secrets
```

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5. Phase 2 — Grant Owner to the new account

```
gcloud projects add-iam-policy-binding musallam-delivery-prod `
  --member="user:NEW_OWNER@example.com" `
  --role="roles/owner"
```

Owner, not Editor. `firebase deploy --only functions` needs Cloud Run, Service Usage, IAM, Artifact Registry and Secret Manager together; Editor alone fails partway through a deploy with `PERMISSION_DENIED`.

Verify both accounts appear under `roles/owner`:

```
gcloud projects get-iam-policy musallam-delivery-prod `
  --format="table(bindings.role,bindings.members)"
```

6. Phase 3 — Billing

A billing account belongs to a Google **payments profile** and cannot be transferred between Google accounts. Only the project's link can move, which unlinks the old account. No money is transferred.

Choose one

A1 — keep the existing billing account, swap the payment method. Preserves invoice history, promotional credits and any Committed Use Discounts.

Console → Billing → Account management:

1. Add the new owner's card as a payment method.
2. Set it as default.
3. Remove the old card.
4. Add the new owner as `roles/billing.admin` on that billing account.

A2 — create a new billing account and relink the project. Clean legal ownership, but credits, CUD and history start from zero.

```
gcloud beta billing accounts list
gcloud beta billing projects link musallam-delivery-prod `
  --billing-account=NEW_BILLING_ID
gcloud beta billing projects describe musallam-delivery-prod
```

Confirm `billingEnabled: true` afterwards.

What happens to the money either way

Item	Outcome
Usage accrued this cycle	Billed on the **old** billing account's next invoice
Issued invoices (paid or unpaid)	Remain on the old account permanently
Prepaid, promotional or free-trial credits	Bound to the old account — **lost**, not moved
Committed Use Discounts	Bound to the old account — **lost**
Payment method	Must be added fresh to the new billing account
Budget alerts and spend notifications	Not carried over — recreate them
Tax documents and GST invoices	Remain on the old account

Timing

Switch on the **1st of a month**. A mid-month relink produces prorated invoices across two billing accounts, which makes reconciliation painful and hides anomalies.

Do not close the old billing account until one full invoice cycle has passed.

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7. Phase 4 — Essential Contacts

Google sends suspension, quota and outage notices to Essential Contacts, not to the project's IAM owners.

```
gcloud essentialcontacts contacts create `
  --project=musallam-delivery-prod `
  --email=NEW_OWNER@example.com `
  --language=en `
  --notification-category=ALL
```

Remove the old contact only after the new one is confirmed receiving mail.

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8. Phase 5 — Prove the new account has full access

Run as the new owner, in a fresh shell.

```
gcloud auth login
gcloud config set project musallam-delivery-prod
gcloud projects describe musallam-delivery-prod
firebase login:add
firebase use musallam-delivery-prod
firebase projects:list
firebase functions:list
firebase deploy --only firestore:rules
```

`firebase functions:list` should return roughly two hundred functions in `me-central1`. The rules deploy is an intentional no-op: if it succeeds, write access is proven. If it fails, return to Phase 2 — do not proceed to credential rotation.

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9. Phase 6 — Rotate credentials

Transferring ownership without rotating leaves the old account fully functional through cached keys. This phase is the actual security work.

9.1 Admin SDK key

```
gcloud iam service-accounts keys list `
  --iam-account=firebase-adminsdk-xxxxx@musallam-delivery-prod.iam.gserviceaccount.com
```

Create a new key, update Vercel `FIREBASE_SERVICE_ACCOUNT_JSON` (or `FIREBASE_CLIENT_EMAIL` plus `FIREBASE_PRIVATE_KEY`), redeploy, verify a production write succeeds, and only then delete the old key.

9.2 Functions secrets

These apply on the next functions deploy, not immediately.

```
firebase functions:secrets:set WORKER_SHARED_SECRET
firebase functions:secrets:set SERVICE_ODOMETER_APP_SECRET
```

9.3 Cloudflare Worker

`infra/workers/dpd-live` calls the functions base URL with an `X-Worker-Secret` header validated against `WORKER_SHARED_SECRET`. Rotate both sides in the same window or the live map stops updating.

```
npx wrangler secret put WORKER_SHARED_SECRET --config infra/workers/dpd-live/wrangler.jsonc
```

9.4 Bites cross-project grant

`functions/src/core/bites.ts` documents that the delivery project's runtime service account needs `roles/datastore.user` on the Bites project. Without it every milestone write returns `PERMISSION_DENIED` and the customer app stalls on "Preparing". Confirm the grant still exists after rotation.

9.5 Vercel

Rotating `CRON_SECRET` requires a redeploy so the crons defined in `vercel.json` pick it up. The project's own Firebase env vars do not change under Option A.

9.6 Confirm nothing in the repository must change

Option A leaves all of these valid as-is:

`.firebaserc`, `firebase.json`, `functions/src/core/fs.ts`, `src/lib/firebase/admin.ts`, `src/lib/firebase/db.ts`, `infra/workers/dpd-live/wrangler.jsonc`, `infra/workers/dpd-live/src/env.ts`, `infra/config.ts`, `infra/Pulumi.production.yaml`, `docs/firebase-prod/*`, `.env.example`.

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10. Phase 7 — Demote the old account

Wait until Phase 5 and Phase 6 are both verified. Then downgrade rather than remove outright:

```
gcloud projects add-iam-policy-binding musallam-delivery-prod `
  --member="user:OLD_OWNER@example.com" --role="roles/viewer"
gcloud projects remove-iam-policy-binding musallam-delivery-prod `
  --member="user:OLD_OWNER@example.com" --role="roles/owner"
```

Keep Viewer for fourteen days so anything overlooked is still recoverable, then remove it entirely. Never remove the last owner, and never remove both owners in the same change.

11. Phase 8 — Verification

```
firebase functions:list
firebase firestore:indexes
firebase auth:export staff-after.json
```

Every composite index must report `READY`; the staff export count and `customAttributes` must match the Phase 1 snapshot.

Then exercise the running system in this order, because each step proves a different layer:

1. Staff login at `dpdadmin-prod.vercel.app` — the sidebar renders, which proves `admin_permissions` and the custom claims survived.
2. `/drivers` loads — Firestore read path.
3. One callable write — proves the `allUsers → roles/run.invoker` binding and the new service account.
4. Rider app login by employee id and passcode — proves Auth was untouched.
5. One FCM push — proves sender id `579224507592` and the token collection are intact.

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12. Rollback

Failure point	Action
Anything breaks after Phase 2	`gcloud projects set-iam-policy musallam-delivery-prod iam-before.json`
Billing relink caused an unexpected charge	Re-link the recorded old billing account id
Deploy fails for the new account	A permission is missing from the Owner grant — compare policies. Do **not** remove the old owner yet
New secret breaks the live map	Restore the previous `WORKER_SHARED_SECRET` on both the Worker and the functions

13. Checklist

- [ ] `iam-before.json`, `functions-before.txt`, `indexes-before.txt`, `staff-before.json` captured
- [ ] Current billing account id recorded
- [ ] New account holds `roles/owner` and is visible in the policy
- [ ] Billing decided (A1 keep account, or A2 relink) and confirmed `billingEnabled: true`
- [ ] Budget alerts recreated on the new billing account
- [ ] Essential Contact created for the new account
- [ ] `firebase deploy --only firestore:rules` succeeds as the new account
- [ ] Admin SDK key rotated, Vercel updated, production write verified, old key deleted
- [ ] `WORKER_SHARED_SECRET` rotated in Secret Manager **and** the Cloudflare Worker
- [ ] `SERVICE_ODOMETER_APP_SECRET` re-set
- [ ] Bites `roles/datastore.user` grant confirmed
- [ ] Staff login, driver list, callable write, rider login, FCM push all verified
- [ ] Old account demoted to Viewer and left for fourteen days

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14. Timeline

Phase	Effort
Snapshots and prerequisites	30 minutes
IAM grant and billing	30 to 45 minutes
Access verification	20 minutes
Credential rotation	1 to 2 hours
Demotion and passive monitoring	14 days, unattended

Total hands-on time is roughly half a working day. Downtime is zero, code changes are zero, and no application release is required.